

CALCULUS FOR ENGINEERS

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Exact differential form

$$\frac{\partial f}{\partial y} = -e^x \sin y + xz + \frac{\partial g}{\partial y} = xz - e^x \sin y \implies \frac{\partial g}{\partial y} = 0.$$

$$f(x, y, z) = e^x \cos y + xyz + h(z).$$

$$\frac{\partial f}{\partial z} = xy + \frac{dh}{dz} = xy + z \implies \frac{dh}{dz} = z.$$

$$\text{so } h(z) = \frac{z^2}{2} + C.$$

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Definition

Any expression $M(x, y, z)dx + N(x, y, z)dy + P(x, y, z)dz$ is a **differential form**. A differential form is **exact** on a domain D in space if

$$Mdx + Ndy + Pdz = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz = df$$

for some scalar function f throughout D .

Component test for exactness

The differential form $M(x, y, z)dx + N(x, y, z)dy + P(x, y, z)dz$ is exact on an open simply connected domain if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}.$$

This is equivalent to saying that the field $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ is conservative.

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This is equivalent to saying that the field $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ is conservative.

Example Show that $ydx + xdy + 4dz$ is exact and evaluate the integral $\int_{(1,1,1)}^{(2,3,-1)} ydx + xdy + 4dz$ over any path from $(1, 1, 1)$ to $(2, 3, -1)$.

Divergence of a Vector field

Definition

Let $v(x, y, z)$ be a differentiable vector function, and v_1, v_2, v_3 are components of v . Then

$$\operatorname{div} v = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

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$$\nabla \cdot v = \left[\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k \right] \cdot [v_1 i + v_2 j + v_3 k] = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

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$$v = [x^2, 4y^2, 9z^2] \text{ and } v = [0, \cos xyz, \sin xyz]$$

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Example Show that the vector $\mathbf{V} = (x + 3y)\mathbf{i} + (y - 3z)\mathbf{j} + (x - 2z)\mathbf{k}$ is solenoidal.

Curl of a Vector field

Definition

Let $v(x, y, z) = [v_1, v_2, v_3] = v_1i + v_2j + v_3k$ be a differentiable vector function. Then the **curl** of the vector function v is defined as

$$\operatorname{curl} v = \nabla \times v = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

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Example Show that the vector $\mathbf{V} = (x^2 + xy^2)\mathbf{i} + (y^2 + x^2y)\mathbf{j}$ is irrotational.

Green's Theorem in the plane

Definition

The **circulation density** of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ at the point (x, y) is the scalar expression

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}.$$

This expression is also called **the k-component of the curl**, denoted by $(\text{curl } \mathbf{F}) \cdot \mathbf{k}$.

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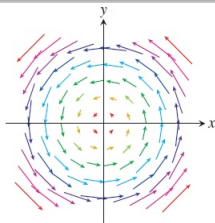
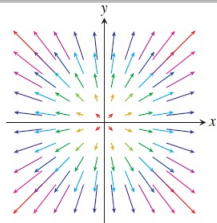
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Example The following vector fields represent the velocity of a gas flowing in the xy -plane. Find the circulation density of each vector field and interpret its physical meaning figure displays the vector fields.

(a) Uniform expansion or compression: $\mathbf{F}(x, y) = cx\mathbf{i} + cy\mathbf{j}$

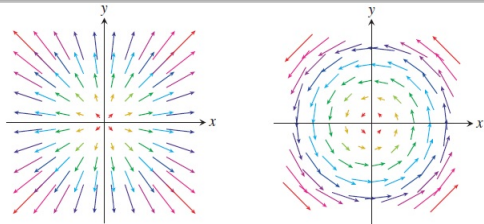
(b) Uniform rotation: $\mathbf{F}(x, y) = -cy\mathbf{i} + cx\mathbf{j}$

Green's Theorem in the plane



(a) Uniform expansion: $(\text{curl} \mathbf{F}) \cdot \mathbf{k} = \frac{\partial}{\partial k}(cy) - \frac{\partial}{\partial y}(cx) = 0$. The gas is not circulating at very small scales.

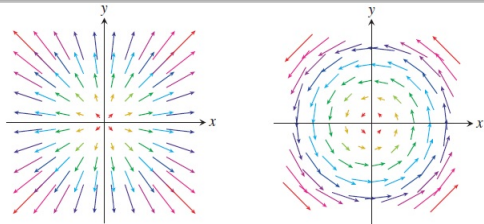
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(b) Rotation: $(\text{curl}\mathbf{F}) \cdot \mathbf{k} = \frac{\partial}{\partial x}(cx) - \frac{\partial}{\partial y}(-cy) = 2c$. The constant circulation density indicates rotation at every point. If $c > 0$, the rotation is counterclockwise; if $c < 0$, the rotation is clockwise.

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Example Find the divergence, and interpret what it means, for the vector field (velocity of a gas flowing in the xy - plane.)

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Solution (a) $\operatorname{div}\mathbf{F} = 2c$; If $c > 0$, the gas is undergoing uniform expansion; if $c < 0$, it is undergoing uniform compression.

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(b) $\text{div}\mathbf{F} = 0$; The gas is neither expanding nor compressing.

Theorem (Green's Theorem (Circulation-Curl or Tangential form))

Let C be a piecewise smooth, simple closed curve enclosing a region R in the plane. Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ be a vector field with M and N having continuous first partial derivatives in an open region containing R . Then the counterclockwise circulation of $\mathbf{F}$ around C equals the double integral of $(\text{curl } \mathbf{F}) \cdot \mathbf{k}$ over R .

$$\oint_C \mathbf{F} \cdot \mathbf{T} ds = \oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Green's Theorem in the plane

Theorem (Green's Theorem (Flux-Divergence or Normal Form))

Let C be a piecewise smooth, simple closed curve enclosing a region R in the plane. Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ be a vector field with M and N having continuous first partial derivatives in an open region containing R . Then the outward flux of $\mathbf{F}$ across C equals the double integral of $\operatorname{div} \mathbf{F}$ over the region R enclosed by C .

$$\oint_C \mathbf{F} \cdot \mathbf{n} ds = \oint_C M dy - N dx = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx dy$$

Green's Theorem in the plane

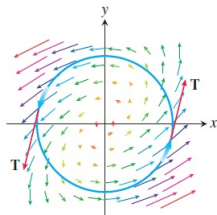
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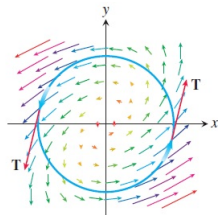
Example Verify both forms of Green's Theorem for the vector field $\mathbf{F}(x, y) = (x - y)\mathbf{i} + x\mathbf{j}$ and the region R bounded by the unit circle $C : \mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}$, $0 \leq t \leq 2\pi$.

Green's Theorem in the plane



$$\begin{aligned}\oint_C Mdx + Ndy &= \int_0^{2\pi} (\cos t - \sin t)(-\sin t dt) + (\cos t)(\cos t dt) = \\ &= \int_0^{2\pi} (-\sin t \cos t + 1) dt = 2\pi \\ \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy &= \iint_R (1 - (-1)) dxdy = 2(\text{area inside the unit circle}) = 2\pi.\end{aligned}$$

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$$\begin{aligned}\text{Again } \oint_C \mathbf{F} \cdot \mathbf{n} ds &= \oint_C M dy - N dx = \\ &= \int_0^{2\pi} (\cos t - \sin t)(\cos t dt) - (\cos t)(-\sin t dt) = \int_0^{2\pi} \cos^2 t dt = \pi. \\ \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx dy &= \iint_R (1 + 0) dx dy = \pi.\end{aligned}$$

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Example Find the area enclosed by the circle

$$\mathbf{r}(t) = (a \cos t)\mathbf{i} + (a \sin t)\mathbf{j}, \quad 0 \leq t \leq 2\pi$$

Curves in plane

Explicit form: $y = f(x)$

Implicit form: $F(x, y) = 0$.

Parametric vector form: $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, $a \leq t \leq b$.

Surfaces and Area

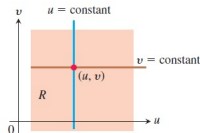
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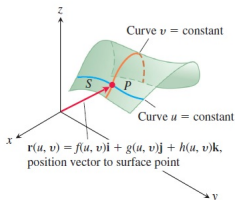
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Surfaces in space



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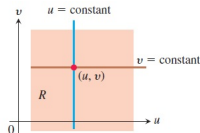
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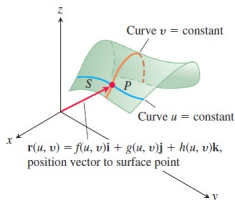
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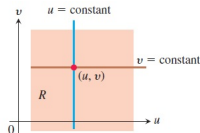
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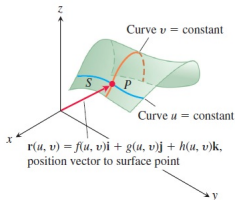
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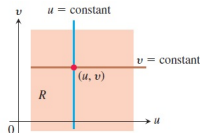
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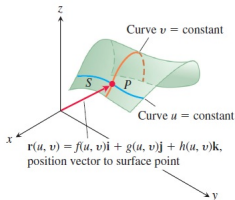
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Solution Here, cylindrical coordinates provides $x = r \cos \theta$, $y = r \sin \theta$, and $z = \sqrt{x^2 + y^2} = r$, $0 \leq r \leq 1$ and $0 \leq \theta \leq 2\pi$.

So $\mathbf{r}(r, \theta) = (r \cos \theta)\mathbf{i} + (r \sin \theta)\mathbf{j} + r\mathbf{k}$, $0 \leq r \leq 1$ $0 \leq \theta \leq 2\pi$.

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$\mathbf{r}(\phi, \theta) = (a \sin \phi \cos \theta)\mathbf{i} + (a \sin \phi \sin \theta)\mathbf{j} + (a \cos \phi)\mathbf{k}$, $0 \leq \phi \leq \pi$, $0 \leq \theta \leq 2\pi$.

Example

Find a parametrization of the cone $z = \sqrt{x^2 + y^2}$, $0 \leq z \leq 1$.

Solution Here, cylindrical coordinates provides $x = r \cos \theta$, $y = r \sin \theta$, and $z = \sqrt{x^2 + y^2} = r$, $0 \leq r \leq 1$ and $0 \leq \theta \leq 2\pi$.

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$x^2 + (y - 3)^2 = 9$, $0 \leq z \leq 5$.

Surfaces and Area

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Example Find a parametrization of the cylinder

$$x^2 + (y - 3)^2 = 9, \quad 0 \leq z \leq 5.$$

Solution $x = r \cos \theta$, $y = r \sin \theta$, $z = z$.

$$x^2 + (y^2 - 6y + 9) = 9 \implies r^2 - 6r \sin \theta = 0.$$

So $r = 6 \sin \theta$, $0 \leq \theta \leq \pi$.

$\mathbf{r}(\theta, z) = (3 \sin 2\theta)\mathbf{i} + (6 \sin^2 \theta)\mathbf{j} + z\mathbf{k}$, $0 \leq \theta \leq \pi$, $0 \leq z \leq 5$.

Surface Area

Definition

A parametrized surface $\mathbf{r}(u, v) = f(u, v)\mathbf{i} + g(u, v)\mathbf{j} + h(u, v)\mathbf{k}$ is **smooth** if $\mathbf{r}_u$ and $\mathbf{r}_v$ is never zero on the interior of the parameter domain.

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Definition

The **area** of the smooth surface

$\mathbf{r}(u, v) = f(u, v)\mathbf{i} + g(u, v)\mathbf{j} + h(u, v)\mathbf{k}$, $1 \leq u \leq b, c \leq v \leq d$ is

$$A = \int \int_R |\mathbf{r}_u \times \mathbf{r}_v| du dv = \int_c^d \int_a^b |\mathbf{r}_u \times \mathbf{r}_v| du dv.$$

Example Find the surface area of the cone $z = \sqrt{x^2 + y^2}$.

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$$|\mathbf{r}_r \times \mathbf{r}_\theta| = \sqrt{2}r.$$

Surface Area

The surface area of the cone is $A = \int_0^{2\pi} \int_0^1 |\mathbf{r}_r \times \mathbf{r}_\theta| dr d\theta = \pi\sqrt{2}$ units squared.

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$\mathbf{r}_\phi \times \mathbf{r}_\theta = (a^2 \sin^2 \phi \cos \theta)\mathbf{i} + (a^2 \sin^2 \phi \sin \theta)\mathbf{j} + (a^2 \sin^2 \phi \cos \phi)\mathbf{k}$

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The area of sphere is $\int_0^{2\pi} \int_0^\pi a^2 \sin \phi d\phi d\theta = \int_0^{2\pi} 2a^2 d\theta = 4\pi a^2$ units squared.

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Definition (Surface area of an Implicit surface)

The area of the surface $F(x, y, z) = c$ over a closed and bounded plane region R is

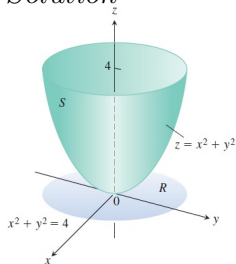
Surface area = $\int \int_R \frac{|\nabla F|}{|\nabla F \cdot \mathbf{p}|} dA$, where $\mathbf{p} = \mathbf{i}, \mathbf{j}$ or $\mathbf{k}$ is normal to R and $\nabla F \cdot \mathbf{p} \neq 0$.

Example Find the area of the surface cut from the bottom of the paraboloid $x^2 + y^2 - z = 0$ by the plane $z = 4$.

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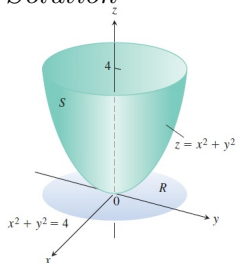
Solution



Surface Area

Example Find the area of the surface cut from the bottom of the paraboloid $x^2 + y^2 - z = 0$ by the plane $z = 4$.

Solution



$$\mathbf{F}(x, y, z) = x^2 + y^2 - z, \quad \nabla \mathbf{F} = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k} \\ \Rightarrow |\nabla \mathbf{F}| = \sqrt{4x^2 + 4y^2 + 1}, \text{ and } |\nabla \mathbf{F} \cdot \mathbf{p}| = |\nabla \mathbf{F} \cdot \mathbf{k}| = 1.$$

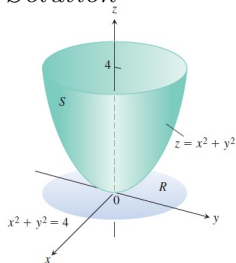
Surface area=

$$\begin{aligned} \iint_R \frac{|\nabla \mathbf{F}|}{|\nabla \mathbf{F} \cdot \mathbf{p}|} dA &= \iint_{x^2+y^2 \leq 4} \sqrt{4x^2 + 4y^2 + 1} dx dy \\ &= \int_0^{2\pi} \int_0^2 \sqrt{4r^2 + 1} r dr d\theta = \\ &= \int_0^{2\pi} \left[\frac{1}{12} (17^{3/2} - 1) \right] d\theta = \frac{\pi}{6} (17\sqrt{17} - 1). \end{aligned}$$

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$$\begin{aligned}\mathbf{F}(x, y, z) &= x^2 + y^2 - z, \quad \nabla \mathbf{F} = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k} \\ \implies |\nabla \mathbf{F}| &= \sqrt{4x^2 + 4y^2 + 1}, \text{ and } |\nabla \mathbf{F} \cdot \mathbf{p}| = |\nabla \mathbf{F} \cdot \mathbf{k}| = 1.\end{aligned}$$

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Definition

For a graph $z = f(x, y)$ over a region R in the xy -plane, the surface area formula is $A = \iint_R \sqrt{f_x^2 + f_y^2 + 1} dx dy$.

Surface Integral of a scalar function

1. For a smooth surface S defined parametrically as

$\mathbf{r}(u, v) = f(u, v)\mathbf{i} + g(u, v)\mathbf{j} + h(u, v)\mathbf{k}$ $(u, v) \in R$ and a continuous function $G(x, y, z)$ defined on S , the surface integral of G over S is given by the double integral over R .

$$\int \int_S G(x, y, z) d\sigma = \int \int_R G(f(u, v), g(u, v), h(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| du dv$$

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2. For a smooth surface S given implicitly by $F(x, y, z) = c$, where F is a continuously differentiable function, with S lying above its closed and bounded shadow region R in the coordinate plane beneath it, the surface integral of G over S is given by the double integral over R ,

$$\int \int_S G(x, y, z) d\sigma = \int \int_R G(x, y, z) \frac{|\nabla F|}{|\nabla F \cdot \mathbf{p}|} dA,$$

where $\mathbf{p}$ is a unit vector normal to R and $\nabla F \cdot \mathbf{p} \neq 0$.

3. For a surface given explicitly as the graph of $z = f(x, y)$, where f is a continuously differentiable function over a region R in the xy -plane, the surface integral of G over S is given by the double integral over R ,

$$\int \int_S G(x, y, z) d\sigma = \int \int_R G(x, y, f(x, y)) \sqrt{f_x^2 + f_y^2 + 1} dx dy.$$

Example Integrate $G(x, y, z) = x^2$ over the cone $z = \sqrt{x^2 + y^2}$, $0 \leq z \leq 1$.

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Surface Integrals

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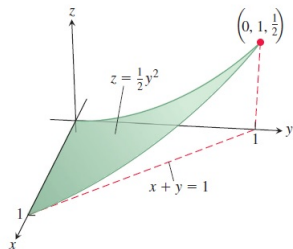
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Example Evaluate $\iint_S \sqrt{x(1+2z)} d\sigma$ on the portion of the cylinder $z = y^2/2$ over the triangular region $R : x \geq 0, y \geq 0, x + y \leq 1$ in the xy -plane.

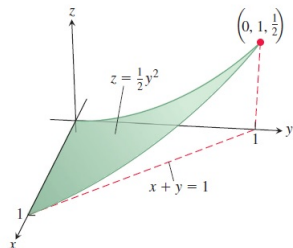
Surface Integrals

Solution



Surface Integrals

Solution



$G(x, y, z) = \sqrt{x(1+2z)} = \sqrt{x}\sqrt{1+y^2}$ and $z = f(x, y) = y^2/2$.

$$d\sigma = \sqrt{f_x^2 + f_y^2 + 1} dx dy = \sqrt{y^2 + 1} dx dy$$

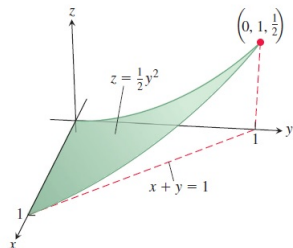
$$\iint_S G(x, y, z) d\sigma =$$

$$\iint_R (\sqrt{x}\sqrt{1+y^2}) \sqrt{y^2+1} dx dy$$

$$\int_0^1 \int_0^{1-x} \sqrt{x}(1+y^2) dy dx = \int_0^1 \sqrt{x}[(1-x) + \frac{1}{3}(1-x)^3] dx = \frac{284}{945} \simeq 0.30$$

Surface Integrals

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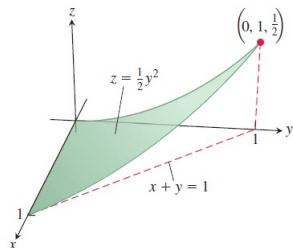
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Note $\iint_S G d\sigma = \iint_{S_1} G d\sigma + \iint_{S_2} G d\sigma + \cdots + \iint_{S_n} G d\sigma.$

Surface Integrals

Solution



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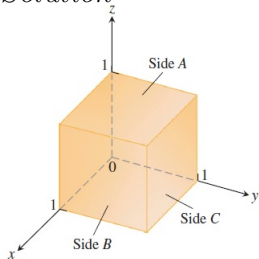
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Note $\iint_S G d\sigma = \iint_{S_1} G d\sigma + \iint_{S_2} G d\sigma + \cdots + \iint_{S_n} G d\sigma.$

Example Integrate $G(x, y, z) = xyz$ over the surface of the cube cut from the first octant by the planes $x = 1, y = 1, z = 1$.

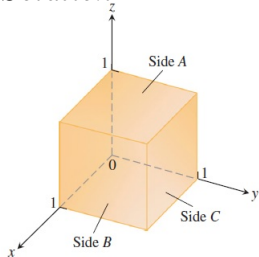
Surface Integrals

Solution



Surface Integrals

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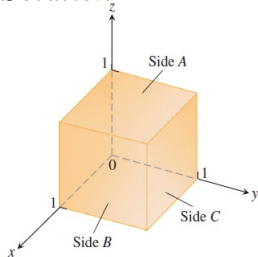
Since $xyz = 0$ on the sides that lie in the coordinate planes, the integral over the surface of the cube

$$\begin{aligned} \iint_{\text{Cube surface}} &= \\ \iint_{\text{Side A}} xyz d\sigma &+ \iint_{\text{Side B}} xyz d\sigma + \iint_{\text{Side C}} xyz d\sigma. \end{aligned}$$

Side A is the surface $f(x, y, z) = z = 1$ over the square region $R_{xy} : 0 \leq x \leq 1, 0 \leq y \leq 1$, in the xy -plane.

Surface Integrals

Solution



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Side A is the surface $f(x, y, z) = z = 1$ over the square region $R_{xy} : 0 \leq x \leq 1, 0 \leq y \leq 1$, in the xy -plane.

$$\text{So } \mathbf{p} = \mathbf{k}, \nabla f = \mathbf{k}, |\nabla f| = 1, |\nabla f \cdot \mathbf{p}| = |\mathbf{k} \cdot \mathbf{k}| = 1$$

$$d\sigma = \frac{|\nabla f|}{|\nabla f \cdot \mathbf{p}|} dA = dxdy.$$

$$xyz = xy(1) = xy$$

$$\int \int_{\text{Side A}} xyz d\sigma = \int \int_{R_{xy}} xy dxdy = \int_0^1 \int_0^1 xy dxdy = 1/4.$$

Due to Symmetry property integrals of xyz over sides B and C are also $1/4$.

$$\text{Hence } \int \int_{\text{Cube surface}} xyz d\sigma = 3/4.$$

Stokes's Theorem

Theorem (Stokes's Theorem)

Let S be a piecewise smooth oriented surface having a piecewise smooth boundary curve C . Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ be a vector field whose components have continuous first partial derivatives on an open region containing S . Then the circulation of $\mathbf{F}$ around C in the direction counterclockwise with respect to the surface's unit normal vector $\mathbf{n}$ equals the integral of the curl vector field $\nabla \times \mathbf{F}$ over S :

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int \int_S \nabla \times \mathbf{F} \cdot \mathbf{n} d\sigma$$

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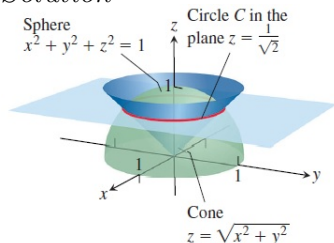
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int \int_S \nabla \times \mathbf{F} \cdot \mathbf{n} d\sigma$$

Example Calculate the circulation of the vector field

$\mathbf{F} = (x^2 + z)\mathbf{i} + (y^2 + 2x)\mathbf{j} + (z^2 - y)\mathbf{k}$ along the curve of intersection of the sphere $x^2 + y^2 + z^2 = 1$ with the cone $z = \sqrt{x^2 + y^2}$ traversed in the counterclockwise direction around the z -axis when viewed from above.

Stokes's Theorem

Solution



The sphere and cone intersect when

$$1 = (x^2 + y^2) + z^2 = 2z^2 \implies z = 1/\sqrt{2}$$

$x^2 + y^2 = 1/2$ and normal vector to the surface $\mathbf{n} = \mathbf{k}$

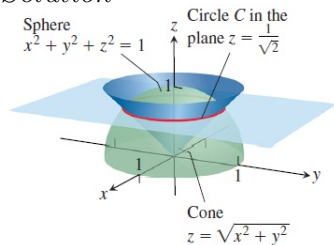
$$\nabla \times \mathbf{F} = -\mathbf{i} + \mathbf{j} + 2\mathbf{k}.$$

so that $\nabla \times \mathbf{F} \cdot \mathbf{k} = 2$ The circulation around the disk is

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \nabla \times \mathbf{F} \cdot \mathbf{k} d\sigma = \iint_S 2 d\sigma = 2 \cdot \text{area of disk} = 2 \cdot \pi \left(\frac{1}{\sqrt{2}}\right)^2 = \pi$$

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Example Calculate the circulation around the bounding circle

$C : x^2 + y^2 = 9, z = 0$ and the field $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$ using the disk of radius 3 centered at the origin in the xy -plane as the surface S .

Solution $\nabla \times \mathbf{F} = -2\mathbf{k}$ and the normal vector $\mathbf{n} = \mathbf{k}$

$$\nabla \times \mathbf{F} \cdot \mathbf{n} d\sigma = -2\mathbf{k} \cdot \mathbf{k} dA = -2dA$$

$$\int \int_S \nabla \times \mathbf{F} \cdot \mathbf{n} d\sigma = \int \int_{x^2+y^2 \leq 9} -2dA = -18\pi.$$

Model question

- 1 A firm's weekly profit in marketing two products is given by $P = 192x + 576y - x^2 - 5y^2 - 2xy - 5000$ where x and y represent the numbers of each units of each product sold weekly. Estimate the average weekly profit if x varies between 40 and 50 units, and y varies between 45 and 50 units.
- 2 Let $f(x, y) = \frac{x-y}{x+y}$, find the directions $\mathbf{u}$ and the values of $D_{\mathbf{u}}f(-\frac{1}{2}, \frac{3}{2})$ is largest.
- 3 Find the work done by the force field $\mathbf{F} = xy\mathbf{i} + (x + y)\mathbf{j}$ on an object moving from the origin $O(0, 0)$ to the point $A(1, 1)$ along the path C , where C is the curve $y = \sqrt{x}$.
- 4 Find the center of mass and the moment of inertia about the y -axis of a thin plate bounded by the x -axis, the lines $x = 1, -1$, and the parabola $y = x^2$ if $\delta(x, y) = 7y + 1$.

Model question

- 1 The electrical charge distribution on a circular plate of radius R meters is $\sigma(r, \theta) = kr(1 - \sin \theta)$ coulomb/m² (k a constant). Find the total charge Q .
- 2 Use Green's Theorem to find the counterclockwise circulation and outward flux for the field $\mathbf{F} = (x - y)\mathbf{i} + (y - x)\mathbf{j}$ and the curve C is the square bounded by $x = 0, x = 1, y = 0, y = 1$.